

Yixuan (Edward) XIONG

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PROFESSIONAL EXPERIENCE

Seattle Office of Emergency Management | Software Engineer Intern (Seattle, WA)

01/2026 – 04/2026

- Served on a 4-person team as technical architect for an **M365** workflow automation platform processing **2,000+ annual requests**; designed a relational schema with event-driven notifications, reducing end-to-end request processing by **83%**.
- Built a multi-criteria instructor matching engine (availability, geographic proximity, language) replacing manual spreadsheets; improved match accuracy from **60% to 80%** and saved **2.5** hours per weekly cycle.
- Implemented a **PII-compliant** data pipeline filtering identifiable fields before publishing aggregated metrics to real-time Power BI dashboards for internal staff and external stakeholders.

Gavel and Quill | Software Engineer Intern (Seattle, WA)

05/2025 - 08/2025

- Built an offline-capable map platform on **AWS EC2** for energy site planning; designed a **PostGIS** spatial database ingesting NOAA and Census datasets, served through **Leaflet-rendered** vector tile layers.
- Engineered a geospatial **ETL** pipeline processing **300+ MB** of daily NOAA data; optimized PostGIS GiST indexes and server-side Vector Tile caching, reducing map query latency by **60%**.
- Implemented row-level **Optimistic Locking** with version counters to handle concurrent data edits during field sync cycles; wrote conflict detection logic that flags and queues conflicting updates for manual review instead of silent overwrites.

Mandarin Oriental Hotel Group | Revenue Operations Engineer (Guangzhou, China)

02/2022 - 02/2023

- Built Python ETL pipelines aggregating booking data from 4 OTA platforms (Ctrip, Expedia, Booking.com, direct channel) into a unified schema; automated daily report generation replacing 8 hours/week of manual Excel consolidation.
- Developed a regression-based demand forecasting model predicting peak booking periods from historical occupancy and seasonal patterns; output fed into a rules-based pricing engine adjusting rates across 3 room categories.

PROJECTS

LeaseLens AI | Agentic Rental Decision Intelligence Platform

06/2026

Python, LangGraph, PostgreSQL, GoogleMaps API, RentCast, FastAPI, Docker

- Reduced rental evaluation effort by **~80%** by consolidating **8+** fragmented housing and neighborhood data sources into an agentic decision pipeline for automated property comparison.
- Designed a **LangGraph-based** multi-agent workflow (Planner, Retrieval, Reasoning, Verifier, Recommendation) to enable stateful task orchestration and evidence-grounded recommendations.
- Mitigated **LLM hallucinations** by grounding responses in normalized API evidence, combining constraint-based filtering with weighted ranking across **20+** rental signals.
- Reduced recommendation latency by **65%** through parallel API orchestration, semantic caching, and asynchronous FastAPI pipelines, minimizing redundant retrieval and LLM inference.

ResponseMatch | LLM Emergency Response Matching System

04/2026

Python, LangGraph, Claude API, ChromaDB, RAGAS, FastAPI

- Self-initiated follow-up to my **Seattle OEM internship**, architected a multi-agent system automating volunteer matching across 40+ Seattle 311 categories, parsing heterogeneous request data into a unified schema (urgency, geo-location, keywords).
- Designed a dual-track validation layer combining LLM semantic matching with rule-based scoring; reduced false-positive matches by 30% vs. LLM-only baseline against a ground-truth set.
- Integrated RAG via ChromaDB grounded on public handbooks; evaluated pipeline using RAGAS framework and LLM-as-Judge, improving Faithfulness metric from 0.62 to 0.84

MaterniFlow | Multi-Agent Clinical Operations System

01/2026

Python, FastAPI, GPT-4 API, PostgreSQL, Next.js

- Designed a **ReAct-based** orchestrator agent that decomposes multi-step clinical requests (e.g. bed transfer + surgery scheduling) into tool calls using GPT-4's function calling API; implemented 3 specialized tool agents with schema-validated parameters.
- Built guardrail middleware intercepting **LLM DB** operations for SQL injection detection and PostgreSQL schema validation, achieving zero unsafe writes across 200+ test queries.
- Trained a logistic regression model on synthetic OB/GYN admission data (1,500 records) to predict patient length-of-stay, integrating predictions into a **FastAPI** endpoint.

TECHNICAL SKILLS

Languages& Frameworks: Python, Java, TypeScript, JavaScript, C/C++, SQL, Spring Boot, FastAPI, React, Next.js, Node.js

Cloud& DevOps: AWS (EC2, Lambda), Azure (Functions, Gov Cloud), Docker, Kubernetes, Nginx, Git, Linux, GitHub CI/CD

Data& AI/ML: PostgreSQL, MySQL, Redis, PostGIS, Pandas, Scikit-learn, LangChain, LangGraph, ChromaDB, RAGAS

EDUCATION

Northeastern University

Seattle, WA

Master of Science in Computer Science

09/2023 - 05/2026

Courses: Database Systems, Distributed Systems, Machine Learning, Cloud Computing, Object Oriented Design

Macau University of Science and Technology

Macau SAR, China

Bachelor of Science

09/2017 - 06/2021